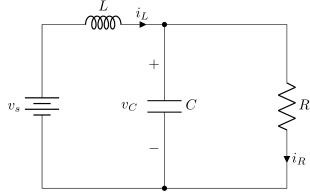


RLC Circuit Simulation and State-Space Modeling

Description

This project demonstrates the simulation of a continuous-time dynamical system using Python. An RLC circuit is modeled using state-space equations and numerically integrated using a forward Euler method. This type of simulation framework forms the basis for more advanced bayesian estimation techniques such as Kalman Filtering.

We will see how we simulate the following RLC circuit:



System Model

The system is modeled as an RLC circuit with the inductor current and capacitor voltage chosen as the state variables:

$$\dot{x} = \begin{bmatrix} \dot{i}_L \\ \dot{v}_C \end{bmatrix}$$

Using standard circuit relationship, the dynamics can be written as

$$\dot{i}_L = \frac{1}{L}(v_s - v_C), \quad \dot{v}_C = \frac{1}{C}(i_L - \frac{v_C}{R})$$

These equations describe the time evolution of the system in terms of the current state and input voltage. For convenience, the system is expressed in compact form as

$$\dot{x} = f(x, u)$$

where $u = v_s$ represents the input to the system.

Numerical Simulation

The continuous-time dynamics are simulated using a forward Euler discretization with a fixed sampling period T . The state is propagated according to

$$x_{k+1} = x_k + T f(x_k, u_k)$$

The simulation is run over a finite time horizon, with the input voltage periodically toggled to excite the system dynamics. At each time step, the current state is stored to enable analysis and visualization of the system behavior over time.

Results

The time histories of the inductor current and capacitor voltage are shown in the figure below. The system exhibits oscillatory behavior due to energy exchange between the inductor and capacitor, with damping introduced by the resistor.

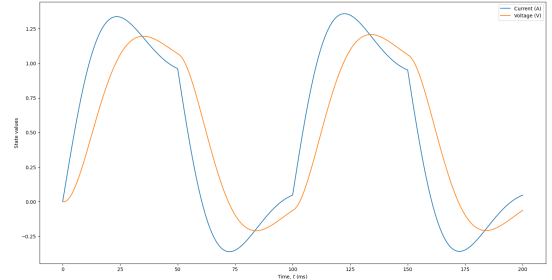


Figure 1: Inductor current and capacitor voltage over time

Next, the resistor and capacitor currents are consistent with the underlying circuit relationships.

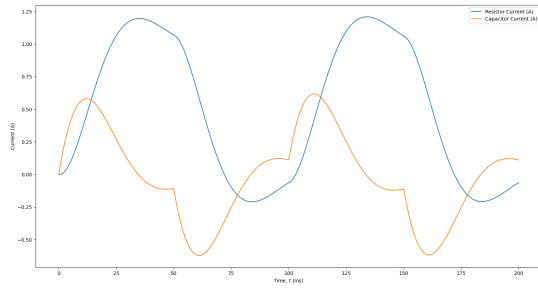


Figure 2: Resistor and capacitor currents over time

Lastly, the equivalent impedance of the circuit. The variation reflects the time-dependent relationship between voltage and current in the system.

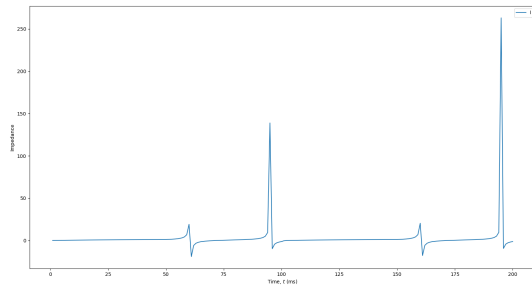


Figure 3: Equivalent impedance over time

Conclusion

This project demonstrates a simple framework for simulating continuous-time dynamical systems using a state-space representation and forward Euler integration. The same structure can be extended to more complex systems and forms the basis for later state estimation methods.